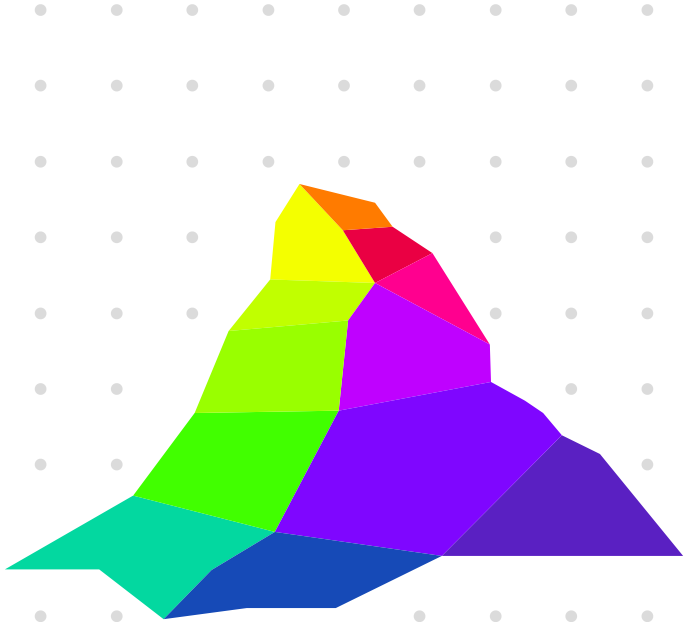


# Mapterhorn Terrain and Imagery

Open Visualization Collaborator Summit 2026

Zürich · Sep 10th, 2026

Oliver Wipfli

# About Me

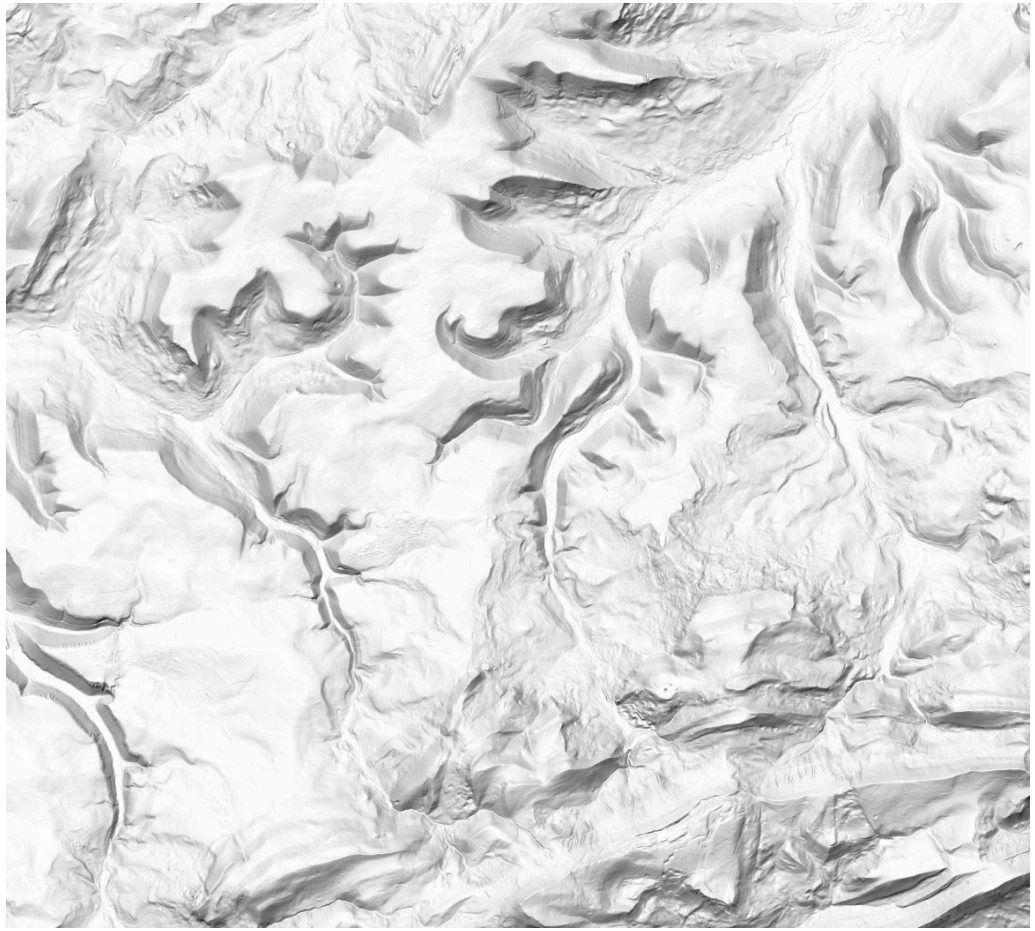
- Oliver Wipfli
- Based in Zürich, Switzerland
- Consultant in Maps and Location Technology
- Email: oliver.wipfli@leichteralsluft.ch
- GitHub: https://github.com/wipfli
- LinkedIn: oliver-wipfli-562258210



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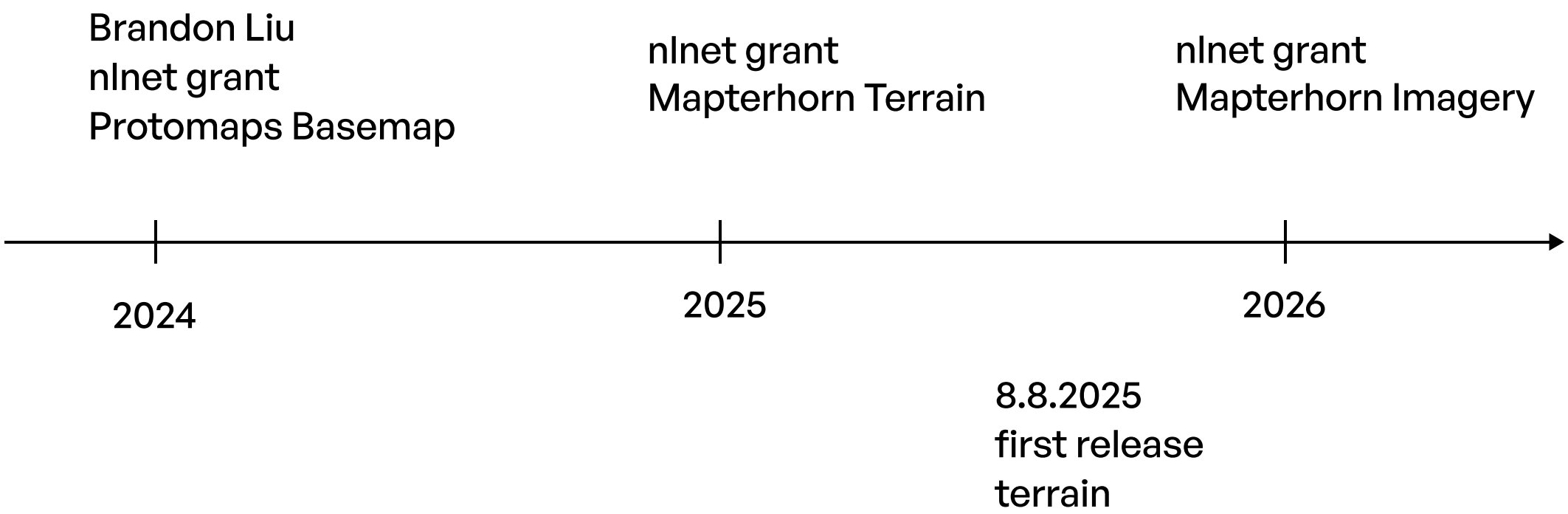
# About Mapterhorn

- Aggregation of open terrain data
- Global low-resolution coverage, local high-resolution lidar
- For hillshading, contour lines, routing, u.s.w.
- Soon also aerial imagery



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# History




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# Outline

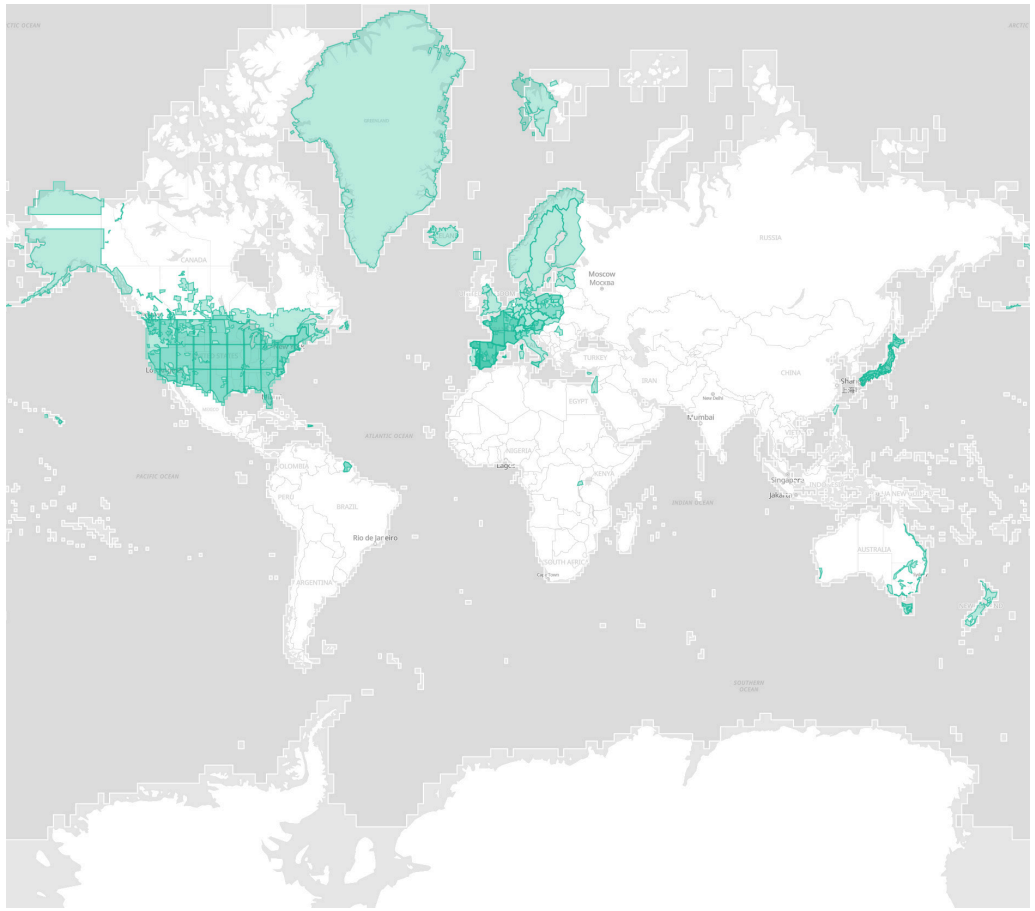
- Input
- Aggregation
- Output and Usage
- Funding and Governance
- Contributing
- Outlook



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# Input

- Global: Copernicus 30 m resolution
- Local: High-resolution Lidar
- 70 data producers
- 18 TiB total input data, shared as tarballs
- mapterhorn.com/attribution
- License CC-BY "or less"

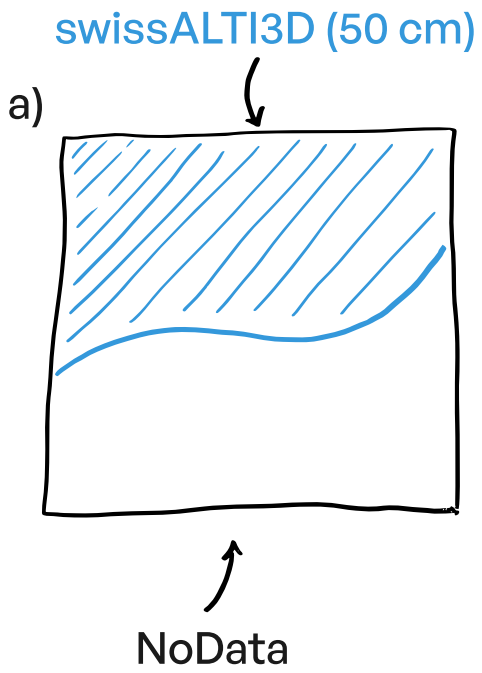



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# Input Normalization

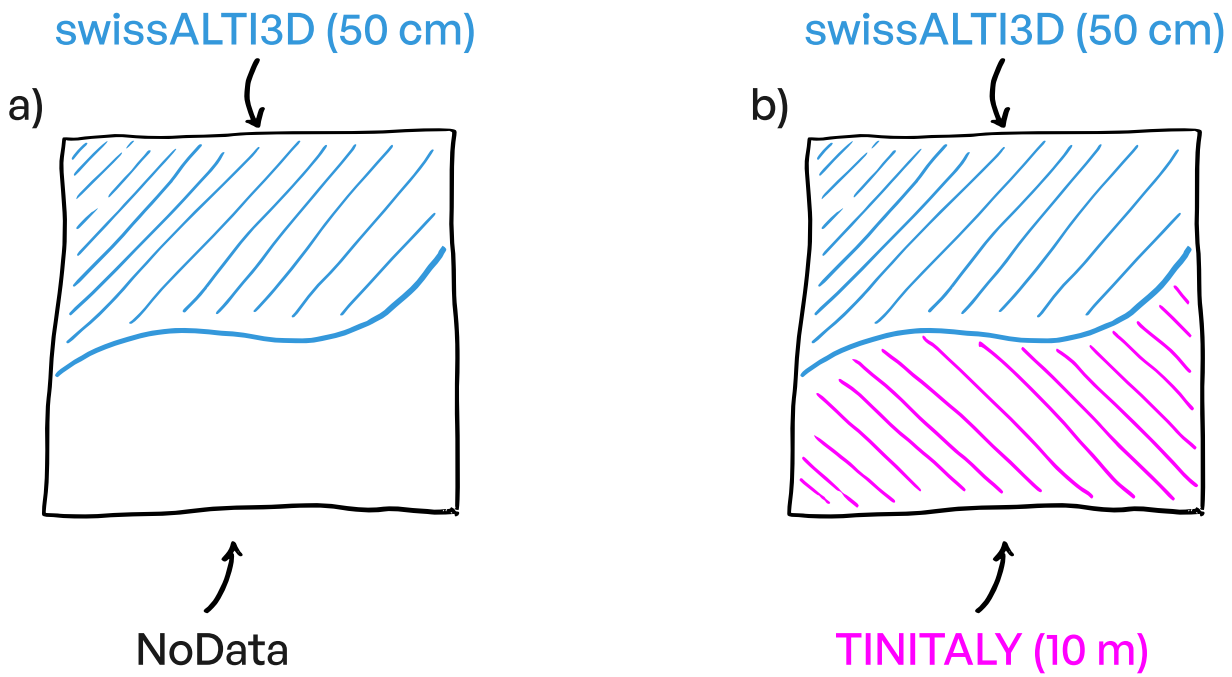
- Download from geoportal
- Store LICENSE.pdf and metadata
- Set consistent projection and NoData value
- Store as GeoTiffs with unified compression and internal tiling
- Pack GeoTiffs, license, and metadata into a tarball
- Normalized tarballs available for download at mapterhorn.com/attribution

# Aggregation



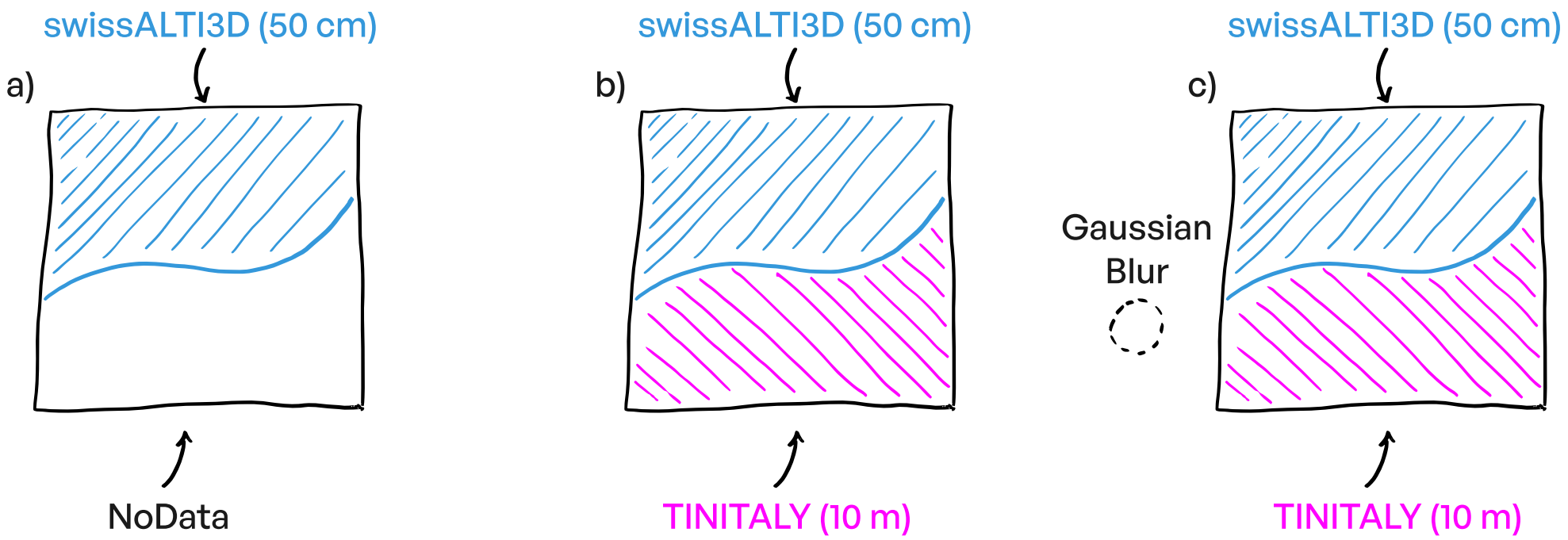

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# Aggregation

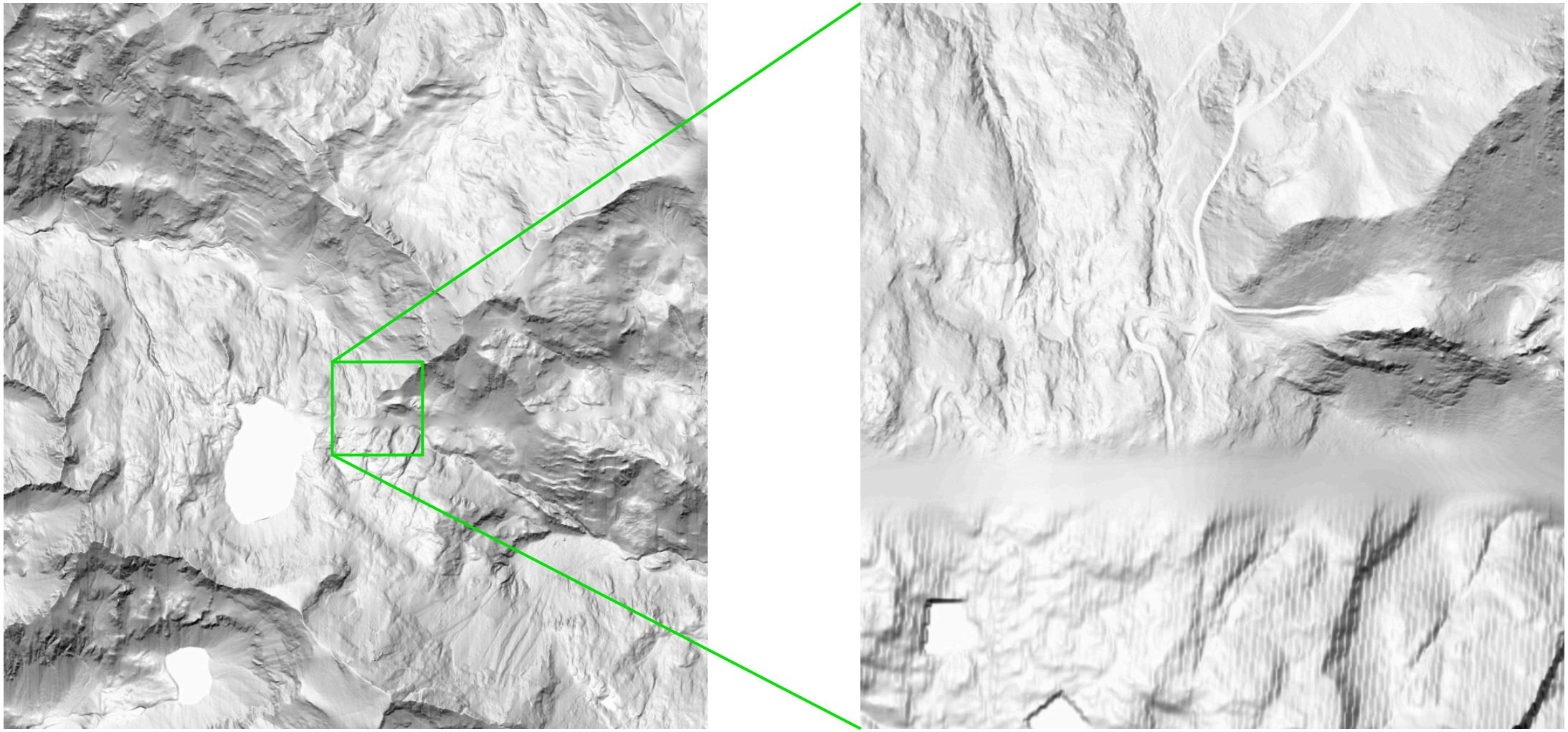



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# Aggregation




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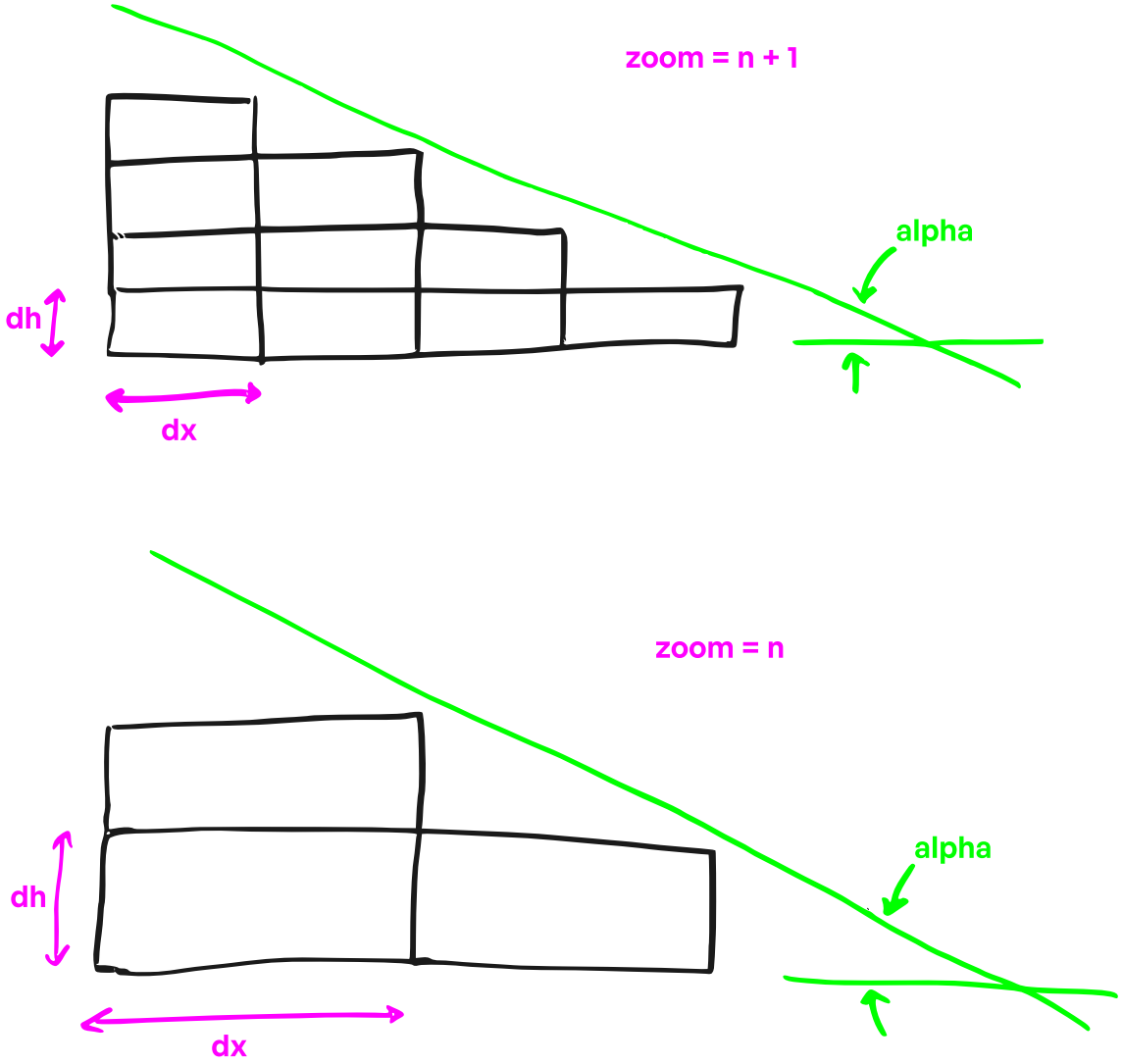
zoom 13.5

zoom 16.5

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# Rounding

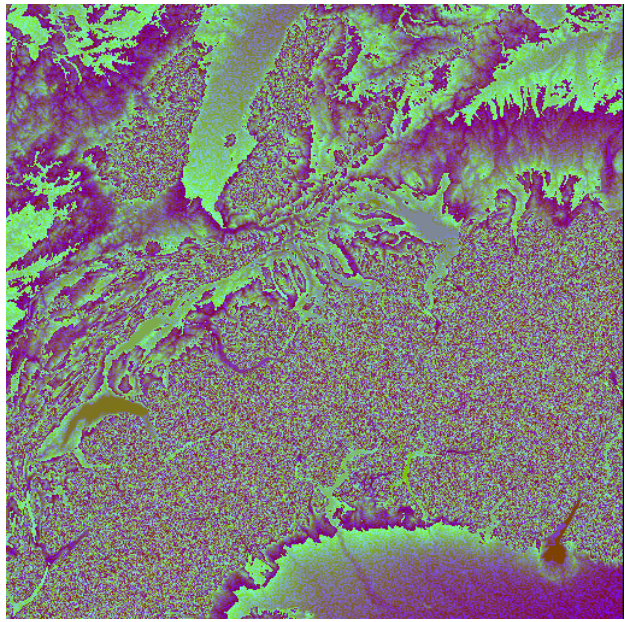
- Terrarium: elevation = 256 * red + green + blue / 256 - 2 ** 15
- zoom 19:
  - dx = 0.15
  - dh = 1 / 256
- zoom 18:
  - dx = 0.3
  - dh = 2 / 256
- zoom 17: ...



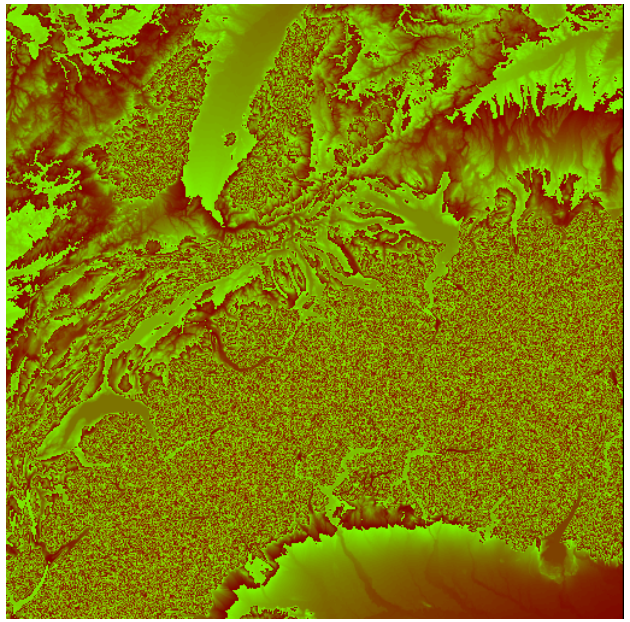

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# Rounding

No Rounding
515 KiB







1 m Rounding
258 KiB



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# Hardware Requirements

## CPU

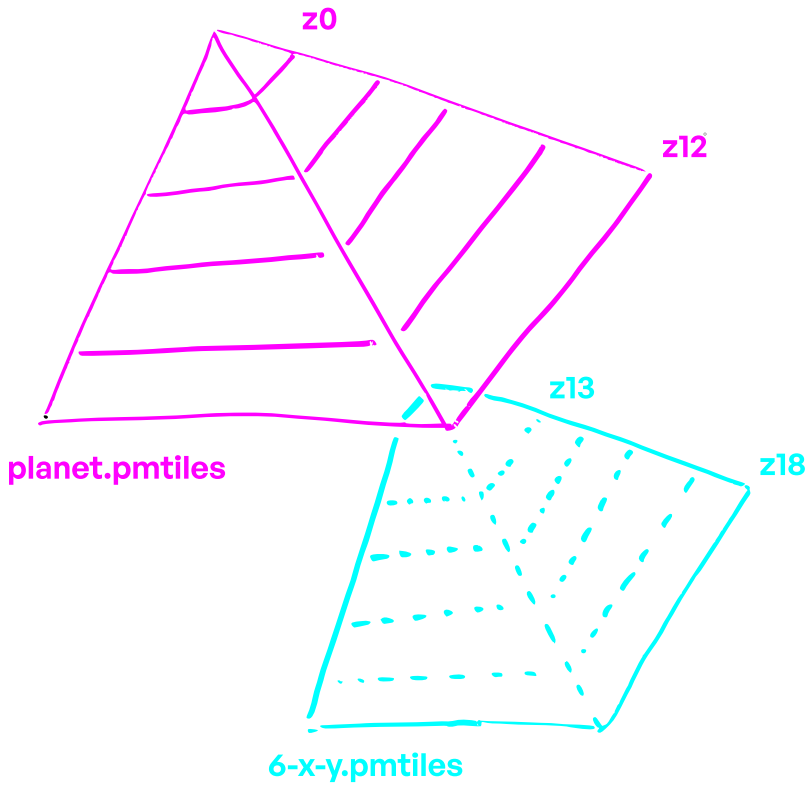
- Total number of tiles: 155M
- WebP lossless encoding ~200 ms per tile -> 359 days
- With 32 threads -> 11 days for WebP encoding alone, add 50% for reprojection and merging

## RAM

- Only 200 MiB per thread -> 6.25 GiB for 32 threads
- Tune down GDAL_CACHEMAX and GDAL_MAX_DATASET_POOL_SIZE

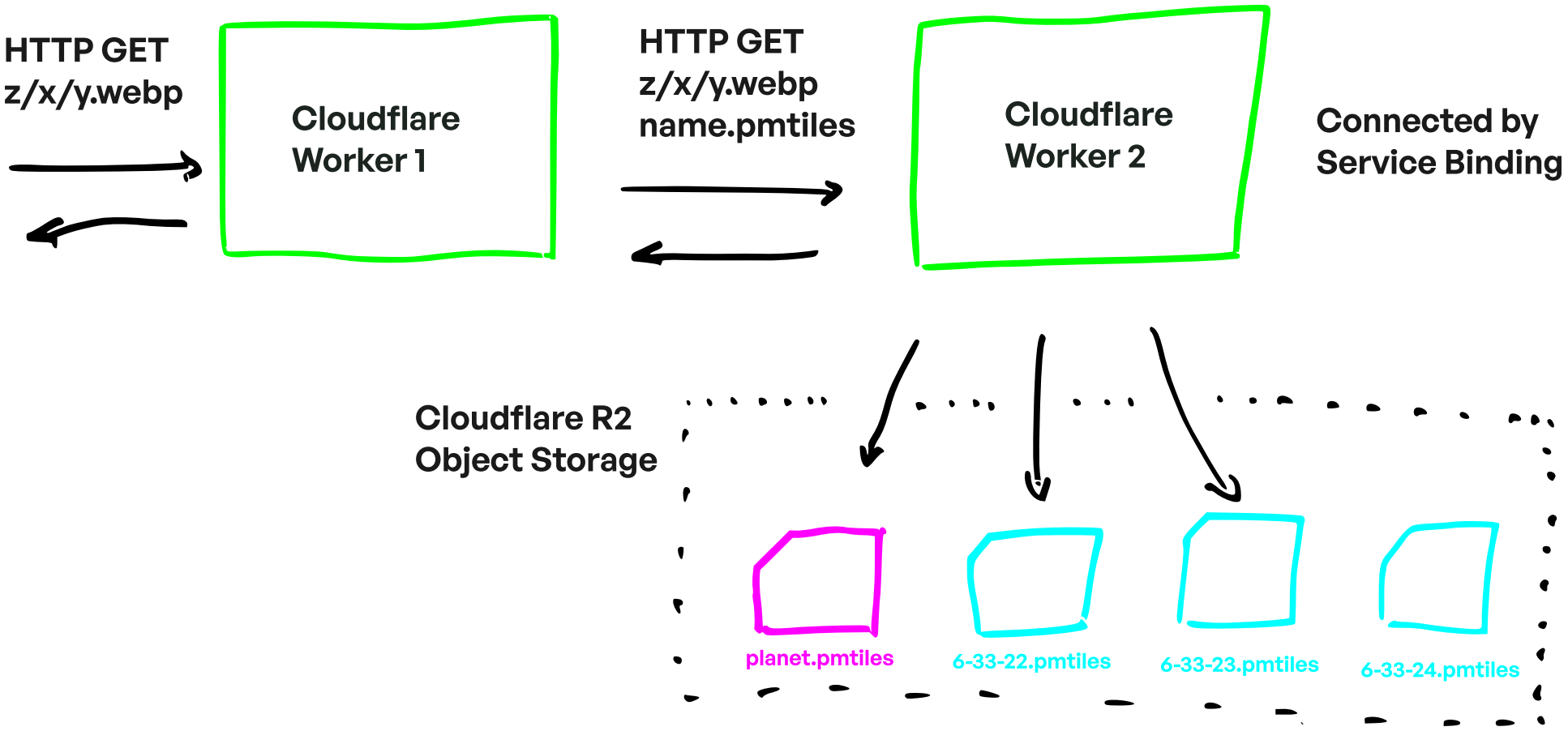
# Output

- 11.8 TiB elevation tiles
- 458 PMTiles archives
- planet.pmtiles: global, zoom 0-12
- 6-x-y.pmtiles: local, zoom 13-var
- mapterhorn.com/data-access




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# Hosting




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# Usage

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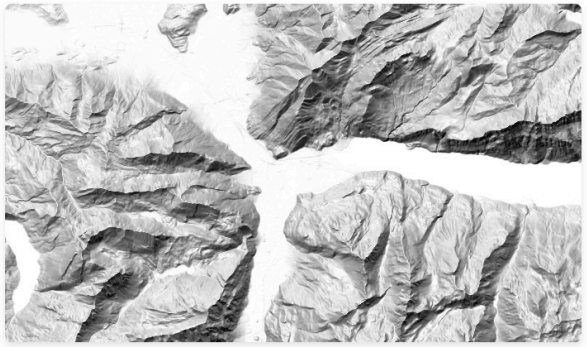
Viewer Data Access Coverage Attribution **Examples**

mapterhorn.com/examples

## Examples

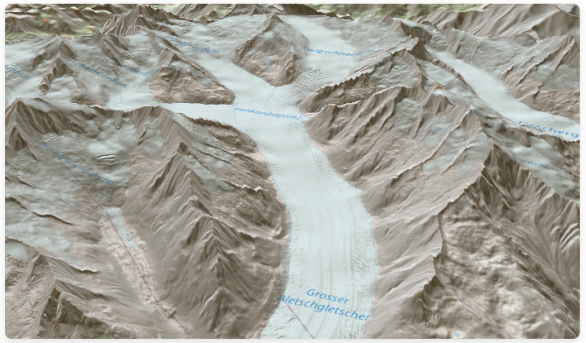
### Hillshade

Add hillshading to a map.



### Terrain

Add 3D terrain and hillshading to an OpenStreetMap raster map.




### Contour

Add contour lines to a map.




### PMTiles

Use Mapterhorn with PMTiles and addProtocol.



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# Users

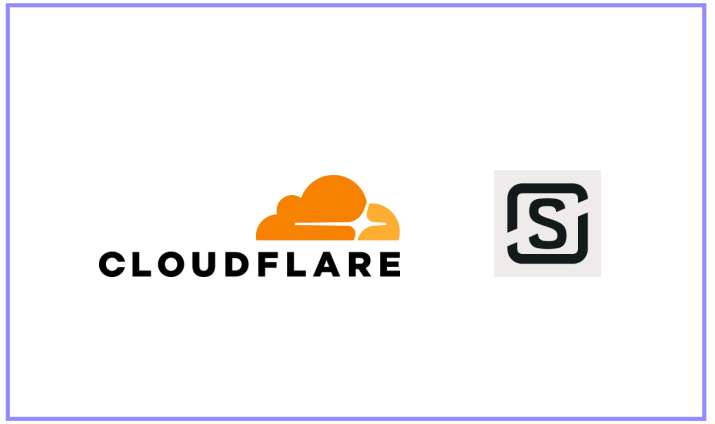
- Media
  - ORF, SVT, zeit.de, FAZ, Seattle Times, The Guardian
- Routing
  - GraphHopper
- Outdoor
  - Komoot, outdooractive

# Funding

opencollective.com/mapterhorn




Research & Development




Web Hosting




Maintenance

Mapterhorn

# Governance



Brandon Liu



Sascha Brawer



Oliver Wipfli

**Full Access**

- GitHub Organization
- Cloudflare Account
- Domain mapterhorn.com
- Financials at OpenCollective

**Fiscal Host**

OpenSourceCollective
501(c)(6) non-profit
440 N Barranca Ave #3939
Covina, CA 91723, USA

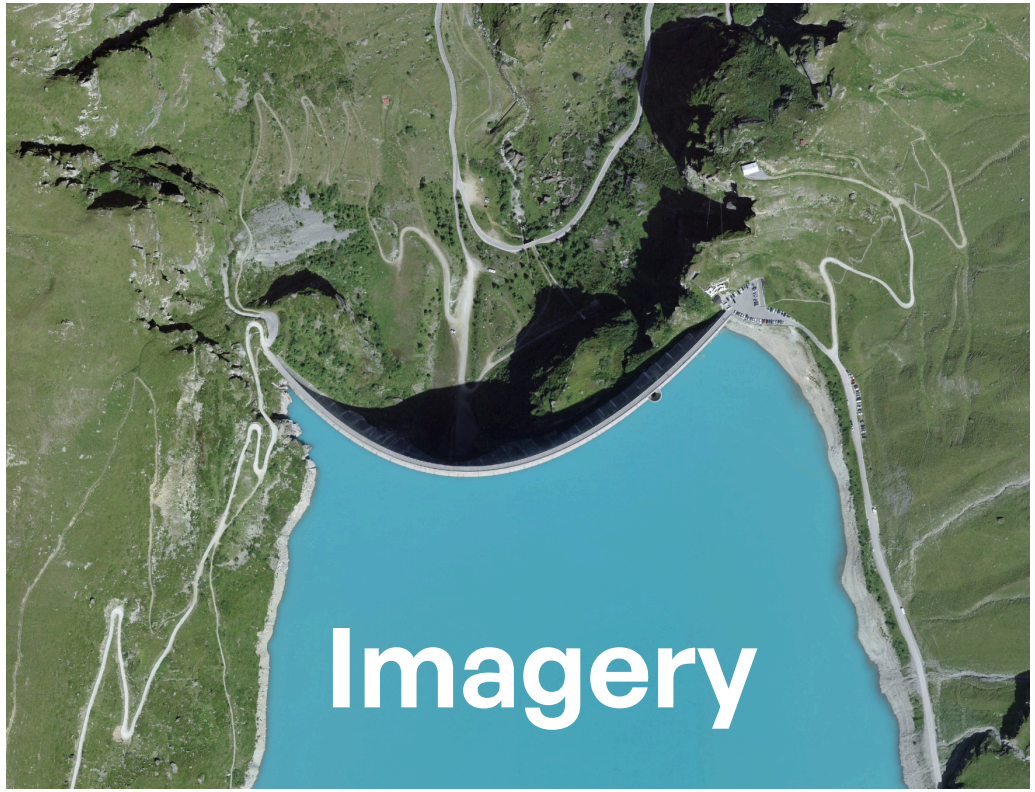
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## Contributing

- Share bug reports, suggest new or updated datasets
  - https://github.com/mapterhorn/mapterhorn
- Contribute financially
  - https://opencollective.com/mapterhorn

# What's Next

- Aerial Imagery
- Same producers, "same" format, same pipeline
- Challenges:
  - 10x more data (100+ TiB)
  - Updates more frequently
- Maintenance, maintenance, maintenance



# Thank You!

- oliver.wipfli@leichteralsluft.ch
- Get a Mapterhorn sticker or key chain after the talk…

